Solution Requirements

Date	15 March 2026
Team ID	xxxxxx
Project Name	Credit Card Approval Prediction
Maximum Marks	4 Marks

Define Functional and Non-Functional Requirements:

Create a comprehensive list of requirements to define exactly what your solution will do and how well it will perform.

- **Functional Requirements (FR)** define specific behaviors, functions, or features of the system (What the system should *do*).
- **Non-Functional Requirements (NFR)** specify the criteria that judge the operation of a system, rather than specific behaviors (How the system should *be*).

Fill out the descriptions and necessary details for each category provided below.

Step 1: Functional Requirements (FR)

S.No	Requirement Category	Requirement Description	Priority (High/Medium/Low)
1	Authentication	*The application allows users to access the prediction form. No login authentication is required for this prototype.*	Low
2	Authorization levels	*The system provides a single user role (Applicant) with permission to enter application details and view prediction results.*	Low

3	External interfaces	The system uses a trained Machine Learning model (`model.pkl`) integrated with a Flask web application.	High
4	Transactions processing	*The system accepts applicant details, preprocesses the data, predicts approval status, and displays the result instantly.*	High
5	Reporting	*The application displays the prediction result (Approved/Rejected) on the result page.*	Medium
6	Business rules, etc.	*Prediction is based on applicant information such as income, employment, education, family status, and other selected features.*	High
7	Compliance to laws or regulations	*The system is intended for educational purposes and should protect user information without storing personal data permanently.*	Medium

8	*Other*	*The application allows users to submit multiple predictions using the "Predict Again" option.*	Medium

Step 2: Non-Functional Requirements (NFR)

S.No	**NFR Category**	**Requirement Description**	**Target Metric / Acceptance Criteria**
1	Performance & Speed	*The application should generate predictions quickly after the user submits the form.*	*Prediction displayed in **less than 2 seconds***
2	Scalability	*The system should support multiple users accessing the application simultaneously.*	*Supports **100+ concurrent users***
3	Security & Data Privacy	*User input should be processed securely and not stored permanently.*	*No permanent storage of personal data*
4	Reliability & Availability	*The application should remain available and provide accurate predictions.*	***99% uptime** on Render deployment*
5	Usability & Accessibility	*The interface should be simple, user-friendly, and easy to navigate.*	*Users can complete a prediction in **less than 2 minutes***

6	*Other*	*The application should be easy to maintain, update, and deploy on different platforms.*	Runs successfully on **Windows** and **Render cloud platform**